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Processing curved surface with a high-speed triple galvanometer laser 3D scanning system

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Abstract: High-speed laser processing of curved surfaces is limited by laser three-dimensional (3D) scanning equipment. It is difficult to balance efficiency and uniformity during processing. In this paper, a self-developed high-speed triple galvanometer laser 3D scanning system (TGLSS) was used. The equations for the laser energy distribution and the spatial spot overlap rate at different positions of the curved surface were derived. The laser interaction mechanism with curved materials and the ablation effect were investigated. Scanning experiments on cylindrical surfaces were conducted to verify the processing differences of the laser at different positions of the curved surface, and the processing quality of the surface was improved by using a variable speed filling method.

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1. Introduction

As an environmentally friendly technology, lasers are widely used in material processing such as microstructure engraving [1–3], coating removal [4–6], drilling [7–9], and texturing [10,11]. To process planar materials, an XY two-dimensional (2D) galvanometer is often used to scan the laser focus in two mutually orthogonal directions (X- and Y-directions) perpendicular to the optical axis. This method allows scanning speeds of up to 10 m/s at a focal length of 160 mm [12]. As production and application requirements escalate to three-dimensional (3D) space, the angle of incidence at which the laser is focused on the curved surface may affect the manufacturing efficiency of the product, the precise shaping of the structure, and the effectiveness of the functional surface [13]. Higher scanning speeds of the laser focus along the optical axis (Z-direction) are required for the study of this effect. The conventional method achieves efficient processing of curved parts by combining a 2D galvanometer with a multi-axis CNC, utilizing the high speed of the galvanometer and the flexibility of a multi-axis system [14,15]. However, the Z-direction focusing capability of the multi-axis CNC is difficult to match the dynamic performance of the galvanometer. This limits the 3D scanning speed of the system and its ability to process curved parts, and limits the study of the scanning process of the laser along the contour of the surface.

The focus scanning speed in the Z-direction can be increased by moving the negative lens along the optical axis, but the Z-focusing speed in this method is largely limited by the speed at which the optics and their accessories can be moved, which is up to three orders of magnitude

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slower than in the X- and Y-directions [16–18]. Manipulating the shape of the lens according to the electrical wetting [19,20] method or the shape-variable lens [21,22] principle allows the Z-direction position of the laser focus to be moved faster. However, the surface shape of a variable lens is difficult to control precisely. Alvarez lenses [23–25] shift focus by moving the position of the lens, but it is difficult to meet the need for continuous and linear focusing. A high-speed triple galvanometer laser 3D scanning system (TGLSS) achieves the same scanning capability in the X-, Y-, and Z-directions using three highly dynamic characteristic galvanometers, as described in our previous work [26,27]. The high-speed TGLSS improves the processing efficiency of curved parts at a similar cost and provides the ability to scan the laser focus directly along the surface contour, which is difficult to achieve at high speeds with conventional systems.

In this paper, a self-developed TGLSS was used, which could increase the laser scanning speed in the Z-direction to the same speed as that in the X- and Y-directions, providing sufficient conditions for realizing the efficient processing of curved parts. Compared with our previous work, we further derive the spot energy distribution of the laser focus on curved surfaces with different incidence angles. Equations for the spatial spot overlap rate at different incidence angles and speeds when scanning along a curved surface contour were derived. The effects of incidence angle and scanning speed on ablation under 3D scanning were verified by simulation and experiment. Based on the spot overlap rate at different incidence angles, a variable-speed laser 3D scanning filling method was developed based on the TGLSS to balance the scanning efficiency and uniformity of curved parts. Parts with cylindrical surfaces were processed using the system, and the efficient 3D scanning capability of the system was verified.

2. Experimental system and materials

2.1. High-speed triple galvanometer laser 3D scanning system

The TGLSS used to perform laser scanning experiments was shown in Fig. 1(a). It consisted of two off-axis parabolic mirrors PM_1 and PM_2 (35588, Edmund), three galvanometers G_1, G_2, and G_3 (ExtraScan II 14, Hans), and an F-theta lens. The PMs had a focal length f of 152.4 mm and an off-axis angle of 45°. The G_1, G_2, and G_3 could be rotated by ±12.5°. The F-theta lens with a focal length f_{FL} of 100 mm was used to focus the laser. A nanosecond fiber laser (YDFLP-100-M8-M-R, JPT) with a beam diameter of 3.0 mm (expanded to 4.5 mm), a wavelength λ of 1064 nm, a beam quality factor M^2 of 1.4, and a maximum average power of 100 W was used for the experiments. The system could scan at speeds of up to 10 m/s in the X-, Y- and Z-directions. As shown in Figs. 1(b) and (c), a cylindrical sample was used to test the focusing effect of the laser scanning along the surface contour. The radius R of the cylindrical surface was 25 mm, and the scanning path was varied along the surface (based on the TGLSS's high-speed Z-focusing capability) with a centroid angle $\Delta\alpha$ of 45°. A round trip fill mode with line spacing s of 200 μm was used.

2.2. Materials and experimental parameters

In the experiments, commercial aluminum films were tested in the experiment by attaching them to the sample, as shown in Fig. (d). In order to investigate the effect of position of the curved surface on the laser processing, orthogonal experiments were carried out using four different incidence angles α: 0°/15°30°/45° and speed from 1 m/s to 5 m/s. The laser scanning path was a rectangle on a cylindrical plane with a line spacing of 200 μm. The experimental laser power was 50 W, and the laser pulse frequency was 150 kHz, and the experimental laser scanning parameters were shown in Table 1. The scanning effects and depth profiles were observed and compared using a laser scanning confocal microscope (VK-X1100, Keyence). Micrographs of sample ablation were observed using a scanning electron microscope (Nova NanoSEM 450, FEI).

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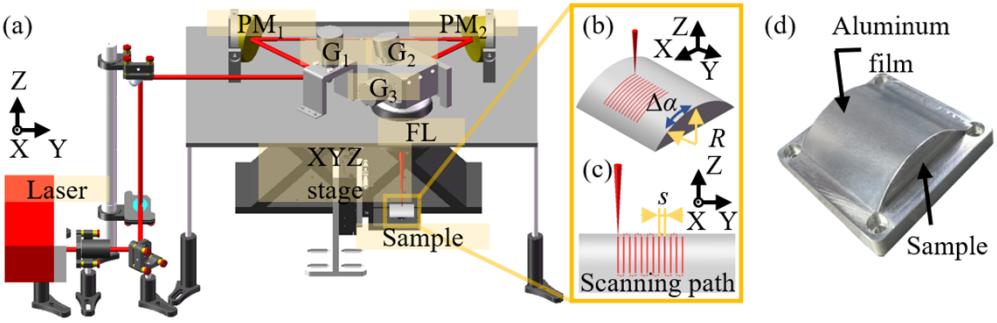


Fig. 1. Diagram of the experiment device. (a) Self-developed high-speed TGLSS; (b) Cylindrical sample; (c) Scanning path (d) Sample with aluminum film

Table 1. Main process parameters of laser scanning

Main parameter	Value
Wavelength λ (nm)	1064
Laser power P (W)	50
Pulse frequency f (kHz)	150
Incidence angles α (°)	0,15,30,45
Speed v (m/s)	1,2,3,4,5
Spot radius w_0 (μm)	19
Line spacing s (μm)	200

3. Analysis and discussion

3.1. Effect of curved parts on laser energy distribution

In space, the angle between the laser beam incident on the surface of a curved surface and the normal to the sample was the angle of incidence α. In order to characterize the energy distribution of the incident laser beam at different positions on the curved surface, the optical axis coordinate system X-Y-Z and the sample coordinate system X_S-Y_S-Z_S are established, as shown in Fig. 2(a), with the Z-axis as the optical axis and the Z_S-axis as the direction normal to the sample. The XY plane was a horizontal plane perpendicular to the Z-axis, and the $X_S Y_S$ plane was the sample surface perpendicular to the Z_S-axis.

For a single mode Gaussian beam, its energy distribution incident on a curved surface could be described by the X_S-Y_S-Z_S coordinate system. We could consider the focused surface area as a plane tangent to the surface, the spot on which the laser was focused was no longer circular but was projected as an ellipse, and its energy distribution could be expressed as Eq. (1) [28].

$$F_S(x_S, y_S, 0) = \frac{2E_S \cos\alpha}{f\pi {w_0}^2 \sqrt{1 + \left(\frac{\lambda x_S \sin\alpha}{\pi {w_0}^2}\right)^2}} \exp\left(-2\left(\frac{{x_S}^2 \cos^2\alpha}{{w_0}^2\left(1 + \left(\frac{\lambda x_S \sin\alpha}{\pi {w_0}^2}\right)^2\right)} + \frac{{y_S}^2}{{w_0}^2}\right)\right) \quad (1)$$

where x_S and y_S were the coordinate values on the X_S-Y_S-Z_S coordinate system, Es was the laser single pulse energy, α was the laser incidence angle, f was the pulse frequency, w_0 was the spot radius, and λ was the laser wavelength.

The energy distributions according to Eq. (1) were shown in Fig. 2(b) (as an example of experimental parameters in Table 1), with the color bar indicating the normalized energy

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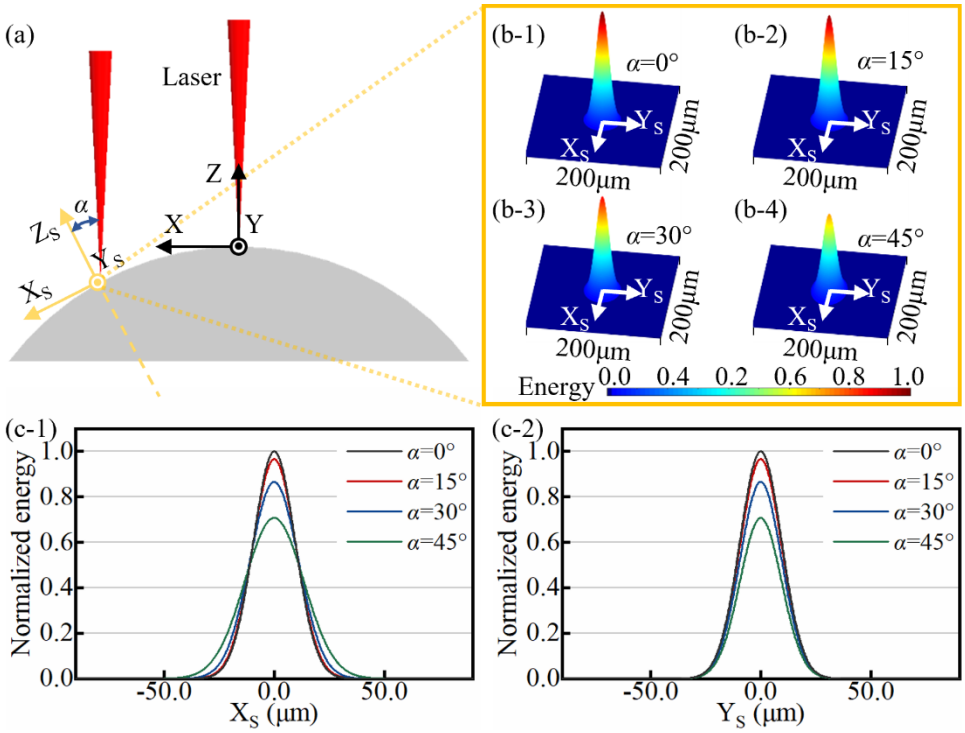


Fig. 2. Distribution of laser energy on a curved surface (a) Definition of incident angle in the X-Y-Z and X_S-Y_S-Z_S coordinate systems; (b) Normalized energy intensity for Gaussian beams at different incidence angles α; (c) Normalized energy intensity in X_S-direction and Y_S-direction

magnitude. The cases at incidence angles $\alpha = 0°$, 15°, 30°, and 45° were represented by Figs. 3(c-1), (c-2), (c-3), and (c-4), respectively. As the incidence angle increases, the peak energy at the center decreased and the concentration of the energy distribution along the X_S-direction decreased, while the concentration of the energy distribution along the Y_S-direction remained constant.

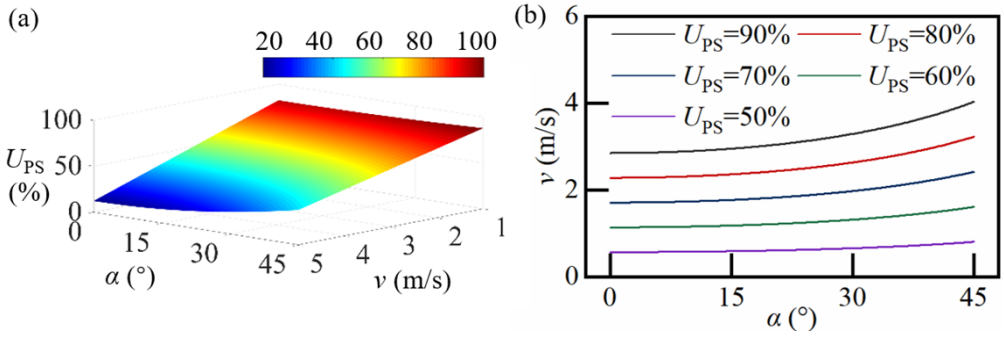


Fig. 3. (a) Variation of the overlap rate U_{PS} of a laser beam incident on a curved surface with α and v; (b) Variation curve between α and v for the same U_{PS}

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3.2. Spot overlap on curved parts

When the Gaussian beam was incident perpendicularly, the overlap rate of laser spots U_P could be expressed as Eq. (2) [29]

$$U_P = (1 - \frac{v}{fd}) \times 100\% \tag{2}$$

where v was the scanning speed, f was the pulse frequency, and d was the length of the laser spot. For a laser beam incident on a curved surface, the length dS of the spot ellipse varied with the angle of laser incidence. The spatial overlap rate of the laser spot U_{PS} could be derived as

$$U_{PS} = (1 - \frac{v \cos\alpha}{2fw_0}) \times 100\% \tag{3}$$

The variation of the U_{PS} of a laser beam incident on a curved surface with α and v (as an example of experimental parameters in Table 1) was shown in Fig. 3(a). At the same scanning speed, the spatial spot overlap increased with the laser incidence angle α, this trend became more obvious as the v increased. When processing curved parts, it was necessary to scan different positions of the curved part uniformly. Consequently, it was necessary to satisfy the same overlap rate at different positions. Figure 3(b) showed the variation curve between incidence angles α and speed v for the same overlap rate U_{PS}.

Part of the data was shown in Table 2. At $v = 1$ m/s, the U_{PS} increased from 82.5% to 87.6% as α increased from 0° to 45°; at $v = 2$ m/s, the U_{PS} increased from 64.9% to 75.2% as α increased from 0° to 45°; at $v = 3$ m/s, the U_{PS} increased from 47.4% to 62.8% as α increased from 0° to 45°; at $v = 4$ m/s, the U_{PS} increased from 29.8% to 50.4% as α increased from 0° to 45°; and at $v = 5$ m/s, the U_{PS} increased from 12.3% to 30.8% as α increased from 0° to 45°. This was due to the fact that the length d_S of the spot along the v-direction increased as α increased. At lower speed ($v < w_0 \times f$), this overlap did not change significantly with α. At higher speed ($v \geq w_0 \times f$), this trend became more obvious.

Table 2. Variation of U_{PS} (%) with α and v for beam incident on a curved surface

	v = 1 m/s	v = 2 m/s	v = 3 m/s	v = 4 m/s	v = 5 m/s
α=0°	82.5	64.9	47.4	29.8	12.3
α=15°	83.1	66.1	49.2	32.2	15.3
α=30°	84.8	69.6	54.4	39.2	24.0
α=45°	87.6	75.2	62.8	50.4	38.0

3.3. Thermal ablation analyses and discussions

According to Eq. (1), it could be derived that when the laser was incident on a point on the curved surface, the energy accumulated at that point on the scanning trajectory was

$$F_S(x_S, y_S, 0) = \sum_{i=-n}^{m} \frac{2E_S \cos\alpha}{f\pi {w_0}^2 \sqrt{1 + \left(\frac{\lambda(x_S + i\frac{v}{f})\sin\alpha}{\pi {w_0}^2}\right)^2}} \exp\left(-2\left(\frac{(x_S + i\frac{v}{f})^2 \cos^2\alpha}{{w_0}^2\left(1 + \left(\frac{\lambda(x_S + i\frac{v}{f})\sin\alpha}{\pi {w_0}^2}\right)^2\right)} + \frac{{y_S}^2}{{w_0}^2}\right)\right) \tag{4}$$

where x_S and y_S were the coordinate values on the X_S-Y_S-Z_S coordinate system, Es was the laser single pulse energy, α was the laser incidence angle, f was the pulse frequency, w_0 was the spot radius, λ was the laser wavelength, v was the scanning speed, m and n were the number of scanning points along the scanning direction and in the opposite direction, respectively.

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The scanning path at different scanning speeds v and different incidence angles α were shown in Fig. 4, and the energy accumulations along the path were normalized. Figure 4 showed that the accumulation of energy decreased with increasing scanning speed at the same α, which was due to the decrease in the number of scanning points on the path. At the lower scanning speeds ($v < 3$ m/s), the distributions of energy along the scanning path did not change obvious with increasing α. At the higher scanning speeds ($v \geq 3$ m/s), the accumulations of energy along the scanning path changed more significantly with increasing α, which was consistent with the conclusions of the derivation of the overlap rate U_{PS} in Section 3.2.

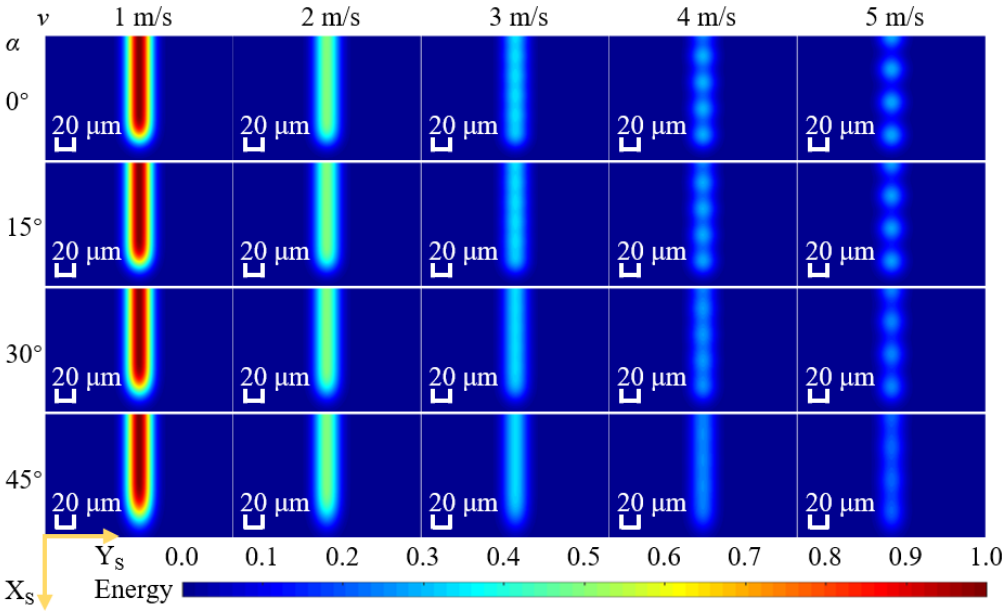


Fig. 4. Simulation results of scanning paths at different α and v

Thermofluidic analysis of laser ablation and molten metal flow on curved surfaces were performed using COMSOL Multiphysics software. Convective heat transfer was considered by combining the unified heat transfer equation with the dynamic mesh method. Pulsed laser ablation of aluminum with scanning speeds v from 1 m/s to 5 m/s were simulated, and the laser incidence angles α varied from 0° to 45° to simulate the laser focusing at different locations on the curved surfaces. The laser energy distributions on the aluminum surface were based on Eq. (1), the scanning parameters were based on Table 1, and the heat transfer equations were as follows:

$$\rho C_p \frac{\partial T}{\partial t} + \rho C_p \mathbf{u} \cdot \nabla T + \nabla \cdot (-k \nabla T) = Q \tag{5}$$

where ρ was the density of aluminum, C_p was the heat capacity, T was the temperature, t was the time, $\mathbf{u}$ was the velocity of the field, Q was the energy input of the laser heat source, and k was the thermal conductivity.

Figure 5 showed the simulation results of scanlines processing with a region of 50 μm selected, in which the angle of incidence was assumed to be constant over this selected region. A lower scanning speed focused more spots on the aluminum, resulting in more heat accumulation and a deeper line depth. This was due to the fact that a larger α resulted in a lower peak energy distribution and a higher spot overlap U_{PS}. The simulation results were consistent with those in Figs. 3 and 4. However, at $\alpha = 45°$, the line depth became shallower because the peak focal energy

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decreased with increasing α and the time that the aluminum surface was above the ablation threshold became shorter.

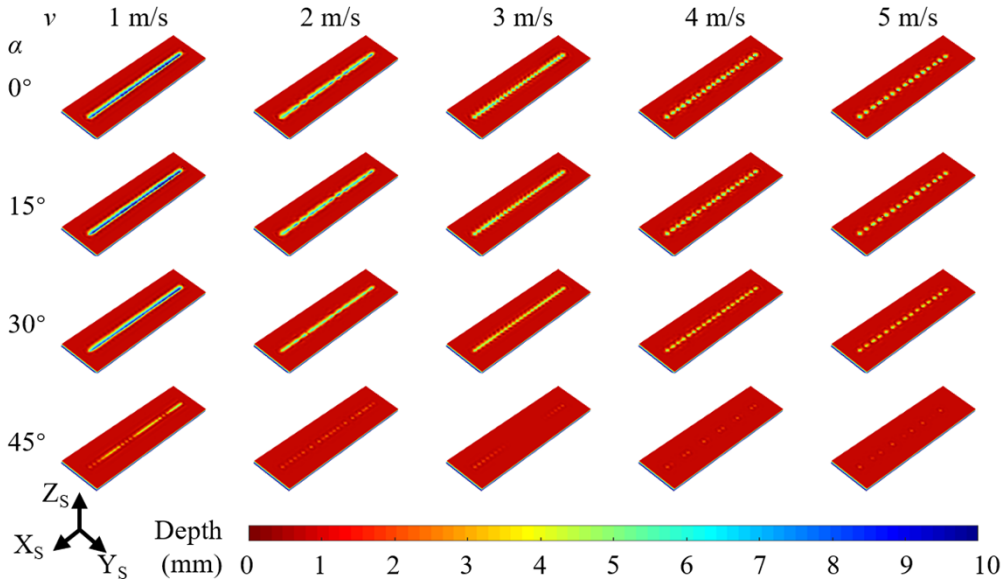


Fig. 5. Simulation analysis results of laser ablation and molten metal flow

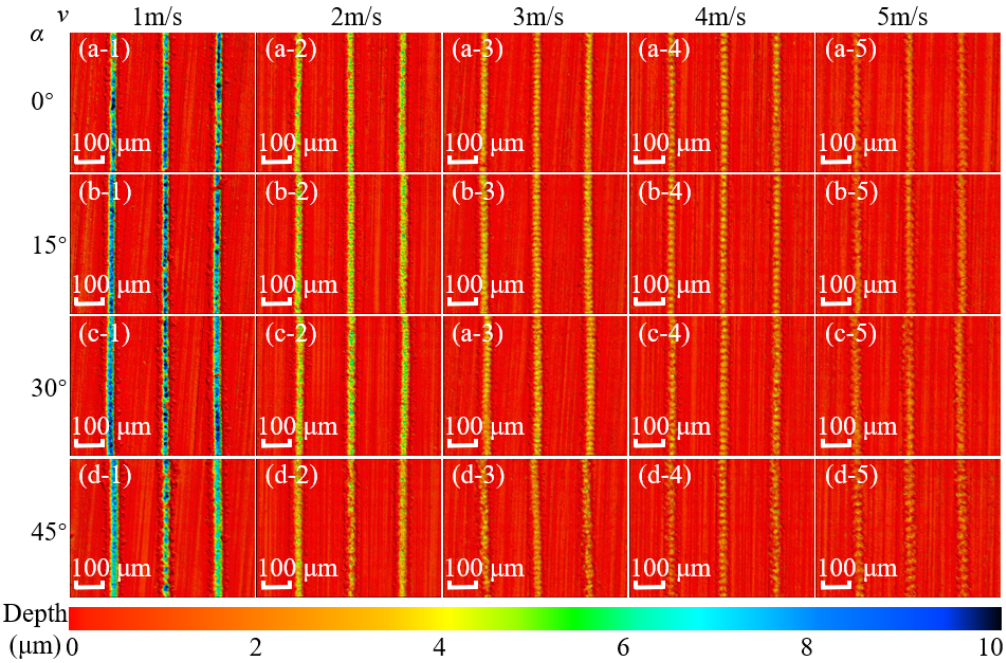


Fig. 6. The results of surface morphology microscopy under laser at different α and v

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4. Results and discussion

4.1. Surface morphology at different incidence angles and scanning speed

The effect of laser beam processing on curved surface morphology was investigated at different angles of incidence and scanning speeds. The experimental parameters were shown in Table 1. The results of surface morphology microscopy under laser at different incidence angles and scanning speeds were shown in Fig. 6. When the angle of incidence α was constant, the depths of the scan line decreased with the increase in v from 1 m/s to 5 m/s. This was due to the decrease in the accumulation time of the laser energy and the number of pulses. As v increased from 1 m/s to 5 m/s, the spot spacing became dispersed from continuous due to a decrease in spot overlap U_{PS}.

When the scanning speed v was constant, the depths of the scan line were almost constant as α increased from 0° to 30°. This was due to the fact that the change in the α did not change the accumulation of the laser energy on the curved surface. At $\alpha = 45°$, the line depth became shallower, which validated the simulation analysis in Fig. 5. The data and variations in specific scan line depths were shown in Fig. 7 and Table 3. In the region of lower scanning speed (v = 1 m/s), the depth of the scan line showed longer error bars. This was due to the higher energy accumulation in this region, which produced more surface oxidation and ablation debris.

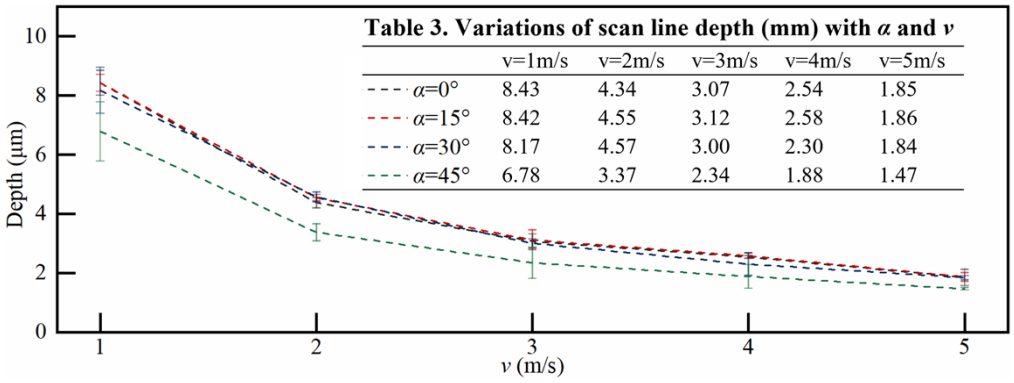


Table 3. Variations of scan line depth (mm) with α and v

	v=1m/s	v=2m/s	v=3m/s	v=4m/s	v=5m/s
α=0°	8.43	4.34	3.07	2.54	1.85
α=15°	8.42	4.55	3.12	2.58	1.86
α=30°	8.17	4.57	3.00	2.30	1.84
α=45°	6.78	3.37	2.34	1.88	1.47

Fig. 7. Variations of scan line depth with α and v

4.2. Microscopic ablation effects at different incidence angles and scanning directions

The magnified SEM micrographs of the ablation effect on curved surfaces by laser beam processing at different incidence angles and scanning speeds were shown in Fig. 8. As shown in Figs. 8(a-1) ~ (a-5), (b-1) ~ (b-5), (c-1) ~ (c-5), and (d-1) ~ (d-5), surface oxides and ablation debris were observed when v = 1 m/s, and the scan line became shallower as the laser scanning speed v increased. At higher magnification (5000 ×) to observe the ablation morphology at the intersection of the focusing spots, it could be observed that the high-pressure steam continued to force the liquid metal at the bottom of the pit to drain around the molten keyhole [30], and the surface oxides and ablation debris became sparse and there were fewer flowing metals and plumes as the scanning speed increased. This was due to the shorter time for energy accumulation.

The reasons for the difference in ablation morphologies with changes in α and v were that higher α increased the spatial spot overlap rate but decreased the energy peak at focusing, resulting in a shorter time for the aluminum surface to be above the ablation threshold. An increase in v decreased the spatial spot overlap and the time for energy to accumulate on the curved surface.

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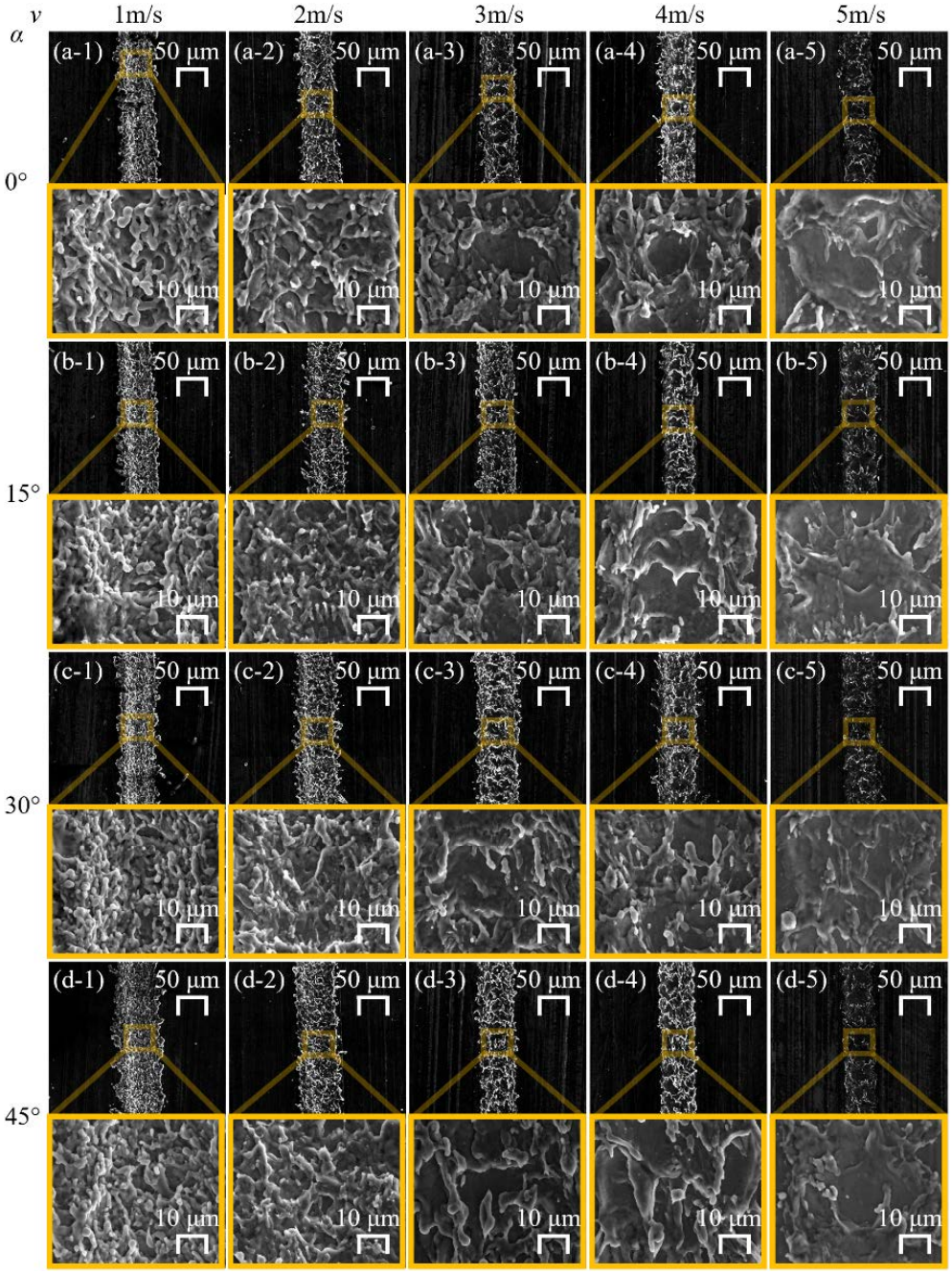


Fig. 8. Magnified SEM micrographs of the ablation effects at different α and v

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4.3. Laser variable speed processing curved surfaces to improve uniformity

As a result of the above analysis, the effect of laser processing on curved surfaces was influenced by the combined effect of the processing position on the curved surface and the scanning speed. By decreasing the scanning speed at the higher incidence speed. By decreasing the scanning speed at higher incidence angle positions, the spot accumulate time in this region could be increased to compensate for the decrease in the peak laser value, thus improving the uniformity of the laser scanning on curved surfaces.

The results and depth profiles of laser scanning on cylindrical samples were shown in Fig. 9. The scanning line spacing was 10 μm and the laser wavelength λ, power P and pulse frequency f were shown in Table 1. As shown in Fig. 9(a-2), when the uniform scanning method was used, the scanning speed v was kept constant at 3 m/s, and the scan line depth decreased with the increase of the laser incidence angle α. As shown in Fig. 9(b-2), when variable speed scanning method was used, the scanning speed v decreased from 3 m/s to 2 m/s with the incident angle α from 0° (top of the cylindrical sample) to 45° (edges of the cylindrical sample), and the scan line depth was almost unchanged with the increase of the α, which indicated that the uniformity of the processing was improved. Compared to the conventional method of rotating the scanning head through a rotary table to compensate for spot shape changes and peak energy reduction, this method was easier, faster. In addition, this method did not have the problem of focus drift caused by the inertia of the rotary table.

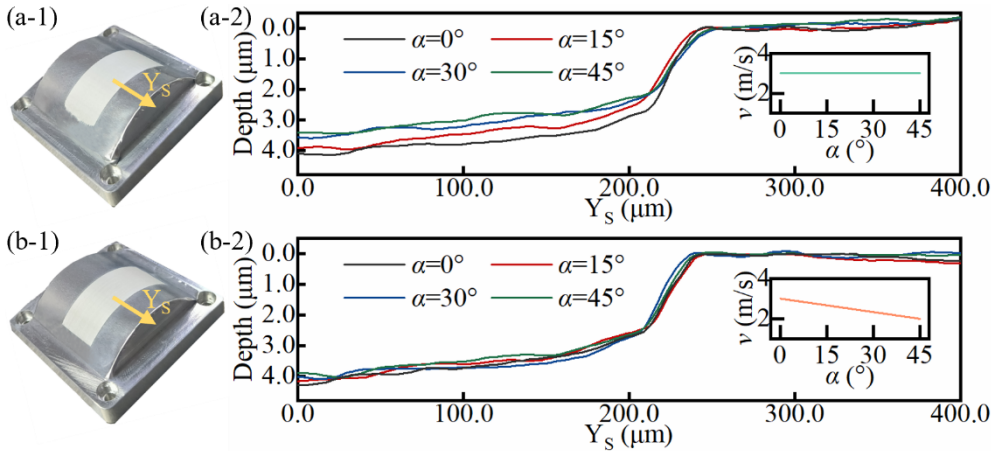


Fig. 9. Results and depth profiles of laser scanning on cylindrical samples (a) Uniform speed method; (b) Variable speed method

5. Conclusion

In this paper, the effect of different positions and different speeds of laser 3D processing of curved surface on infrared nanosecond laser ablation of aluminum was systematically investigated. The energy distribution of the laser incident on a curved surface and the spatial spot overlap rate in scanning along the curved surface contour were derived and modeled. The thermal effects of laser ablation of curved aluminum at different incidence angles and speeds were simulated, experimented and analyzed. The scanning uniformity of the laser on curved materials was improved by developing a filling method with variable speed scanning to compensate for the low peak energy of the laser in regions with high angles of incidence. A reference was provided for the further development of laser technology for efficient and uniform processing of curved surfaces. The technology and equipment were expected to be integrated and applied in aerospace,

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automotive and precision manufacturing [31–33]. The industrialized scanning head can be mounted on the multi-axis mechanism, replacing the conventional laser scanning head (XY galvanometer and Z-dynamic focusing), and improving the efficiency and accuracy of laser 3D scanning processing. The main conclusions were as follows:

(1) Based on the method of approximating smaller curved surface areas as inclined planar areas, the conversion of laser incidence to different positions on the surface to laser incidence on the plane at different angles of incidence. Combined with the energy distribution equations of Gaussian beam, the energy distribution model of the laser incident at different positions of the curved surface was established. As the angle of incidence increased, the peak energy at the center decreased and the concentration of the energy distribution along the direction of the curved surface contour decreased, while the concentration of the energy distribution along the other orthogonal direction remained constant. It should be noted that for high curvature regions, we can further reduce the length of the approximation region to minimize the error and thus improve the overall modeling accuracy.

(2) The equations for the spatial spot overlap rate as a function of incident angle and scanning speed for laser scanning along a curved surface contour were derived. Since the projection of the spot on the surface was elongated with increasing angle of incidence, the spatial spot overlap increased with increasing angle of incidence. This trend changed more significantly at higher scanning speed (> spot radius w0 × pulse frequency f).

(3) The ablation effect of laser scanning at different speeds at different positions on curved aluminum surfaces were simulated by COMSOL software. It was found that the depth of the scan line decreased significantly when the laser scanned the higher incidence angle (α=45°) region of the curved surface, which was due to the decrease of the peak laser energy resulting in a shorter time for the material to be above the ablation threshold. In accordance with the conclusion of (2), the spatial spot overlap rate was higher in the higher incidence angle region, but at $\alpha = 45°$, the increase in the spatial spot overlap rate could not compensate for the decrease in the focal peak energy. The above simulations were confirmed by experiments and microscopic observations.

(4) Summarizing the above analysis, we found that the processing effect of laser on curved surfaces was affected by the combination of incidence angle and scanning speed. Therefore, processing curved samples with conventional uniform scanning speed method inevitably posed problems of processing uniformity. A filling method with variable speed scanning was designed to reduce the scanning speed at the higher incidence angle position of the curved surface to compensate for the decrease of peak energy at the focus. The improvement of uniformity by variable speed scanning was confirmed by performing uniform (speed = 3 m/s) scanning and variable speed scanning (speed changed from 3 m/s to 2 m/s) on cylindrical samples, respectively.

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Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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